



1. What is the primary purpose of a communication link in a network?
 - A) To connect two devices for data transfer
 - B) To increase the power of transmitted signals
 - C) To encrypt the data being sent
 - D) To establish security measures on a network
2. At the network edge, which of the following typically acts as the primary initiators of communication?
 - A) Routers
 - B) Clients
 - C) Servers
 - D) Switches
3. What is the advantage of using a full-duplex communication link?
 - A) It allows data to be transmitted in only one direction at a time.
 - B) It allows data to be transmitted and received simultaneously.
 - C) It reduces the signal strength needed for transmission.
 - D) It decreases the overall bandwidth required for communication.
4. Which of the following is NOT a characteristic of a server in a network?
 - A) It typically provides resources or services to clients.
 - B) It is usually more powerful and capable of handling multiple requests simultaneously.
 - C) It primarily initiates communication with clients.
 - D) It often resides in a data center for better scalability and availability.
5. What is typically housed within a data center?
 - A) Home routers and switches
 - B) High-performance computing devices and large-scale servers
 - C) Personal computers and client devices
 - D) Modems and local network devices
6. What is the primary function of the network core in a large-scale communication network?
 - A) To provide end-user access to network services
 - B) To interconnect various networks and route data between them
 - C) To serve as the physical location for data storage and computing
 - D) To secure network communications and encrypt data packets
7. What is the purpose of interconnected routers in the network core?
 - A) To manage user authentication and data encryption
 - B) To route data packets between different networks based on the destination address
 - C) To provide direct connections to client devices for data transfer
 - D) To generate and manage IP addresses for devices on the network
8. Which of the following physical media is commonly used in fiber-optic communication?
 - A) Copper wires
 - B) Radio waves
 - C) Glass or plastic fibers
 - D) Microwaves
9. Which of the following best describes packet switching?
 - A) A communication method where a dedicated communication path is established between two devices.
 - B) A method of data transmission where messages are broken into small packets and sent independently through the network.
 - C) A technique for transmitting data over a continuous electrical signal.
 - D) A system where a fixed amount of bandwidth is allocated to each user in a network.
10. In circuit switching, the dedicated path between two communicating devices:
 - A) Is established for the duration of the communication session and remains open.
 - B) Is dynamically assigned and terminated at the end of each communication.
 - C) Is shared among multiple users, reducing the need for bandwidth.
 - D) Is only used for short bursts of data transmission.
11. What is the main purpose of HTTP (Hypertext Transfer Protocol)?
 - A) To encrypt data between web browsers and servers
 - B) To transfer files over the Internet
 - C) To enable communication between web browsers and web servers
 - D) To manage email transmission
12. Which HTTP method is used to retrieve data from a specified resource on a server?
 - A) POST
 - B) PUT
 - C) GET
 - D) DELETE
13. What is an HTTP cookie used for?
 - A) To store the IP address of the server
 - B) To store user-specific information for maintaining state between requests
 - C) To track the IP address of the client
 - D) To store the encryption keys for secure communication



14. Which of the following is NOT a feature of IMAP?
 - A) Allows multiple devices to access the same email inbox
 - B) Keeps emails on the server even after they are read
 - C) Deletes emails from the server when they are accessed
 - D) Synchronizes the email status across devices
15. Which of the following protocols is used to receive and manage email on a server, allowing access from multiple devices without losing email data?
 - A) SMTP
 - B) POP3
 - C) IMAP
 - D) FTP
16. When using POP3, what happens when an email is downloaded from the server?
 - A) The email is stored on the server and marked as read
 - B) The email is deleted from the server
 - C) The email is copied to the server for backup
 - D) The email is moved to a folder on the client side only
17. Which email protocol does not allow email management on the server (such as organizing folders or marking emails as read), but instead downloads and removes the email from the server?
 - A) SMTP
 - B) IMAP
 - C) POP3
 - D) HTTP
18. SMTP can use which of the following methods to secure email communication?
 - A) Encryption with SSL/TLS
 - B) Email compression
 - C) Authentication with a username and password
 - D) All of the above
19. Which of the following is a characteristic feature of Peer-to-Peer (P2P) applications?
 - A) They rely on a central server to manage connections between peers.
 - B) They allow direct communication between clients without the need for an intermediary.
 - C) They are primarily used for cloud storage.
 - D) They can only be used for file sharing.
20. In a Peer-to-Peer (P2P) network, what is the primary role of a "peer"?
 - A) To act as a central server for managing network traffic
 - B) To share resources like files, bandwidth, or processing power with other peers
 - C) To monitor the network for security threats
 - D) To host the primary database of the network
21. Which of the following is a key difference between TCP and UDP?
 - A) TCP is connection-oriented, while UDP is connectionless.
 - B) TCP provides faster transmission than UDP.
 - C) UDP guarantees reliable data delivery, while TCP does not.
 - D) TCP is used for streaming media, while UDP is used for email.
22. Which of the following is the primary purpose of the Domain Name System (DNS)?
 - A) To provide encryption for web traffic.
 - B) To translate domain names into IP addresses.
 - C) To ensure secure communication between email servers.
 - D) To store the content of web pages.
23. What is the purpose of a TTL (Time-to-Live) value in DNS records?
 - A) To define how long a DNS record is cached by DNS resolvers.
 - B) To set the expiration time for a domain name registration.
 - C) To specify how long a DNS server should process a request.
 - D) To determine the maximum number of DNS requests allowed in a given period.
24. Which of the following is the primary responsibility of the transport layer in the OSI model?
 - A) Error detection and correction at the bit level
 - B) Reliable end-to-end communication between hosts
 - C) Routing packets across different networks
 - D) Addressing devices on the same network
25. Which of the following protocols operates at the transport layer and ensures reliable data transmission?
 - A) IP
 - B) UDP
 - C) TCP
 - D) ARP
26. Which of the following is a feature of UDP (User Datagram Protocol)?
 - A) Guaranteed delivery of data
 - B) Connection-oriented communication
 - C) Lower overhead and faster transmission compared to TCP
 - D) Error-free communication
27. Which of the following is true about flow control in the transport layer?
 - A) It regulates the rate at which data is sent to prevent overwhelming the receiver.
 - B) It ensures the reliable delivery of packets over the network.